

ADAPTIVE STAIRCASE PROCEDURE

PSY310: Lab in Psychology

Lab Report

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GitHub Link:

Description

This experiment assesses orientation discrimination using the adaptive staircase method, estimating the difference threshold (just noticeable difference) for tilt perception. This experiment is based on the difference threshold, i.e. just noticeable difference(JND). JND represents the smallest angular deviation that the participant can detect easily.

Purpose

The purpose of the experiment was to quantify the perceptual threshold required for reliably distinguishing between left- and right-tilted stimuli. By adjusting stimulus intensity, the adaptive staircase procedure provides an efficient way to estimate the participant's sensitivity to orientation differences. This allows us to measure the discrimination with utmost precision and minimize unwanted trials.

Introduction

The Adaptive Staircase method is used extensively in recent years to determine the signal level corresponding to a particular response's probability. It is a psychophysical method also known as the 'method of ups and downs'. Adaptive methods are designed to concentrate data sampling within a particular topic of the psychometric function.

The Adaptive Staircase was firstly examined by Dixon and Mood(1948). They highlighted that this method required less complex calculations as compared to traditional psychophysical techniques. Since then, the method has been applied in various areas such as vision studies, hearing research, and many more.

Method

The Adaptive Staircase Method was performed by a 21 year female from Ahmedabad University. Her consent was taken first after explaining the experiment and its objectives.

Materials

The hardware which was used was a laptop (MacBook Air) containing a standard keyboard and a 13.3 inch screen display. The resolution was 1000*600.

For the software, an app called PsychoPy was used for crafting the experiment. Microsoft Excel was used to analyse and create visual graphs.

Procedure

The experiment started by creating a fixation for 0.5 seconds.

After that, the figure showing the tilt was displayed. It was a sinusoidal shape with a Gaussian mask, which is shown below (Figure 1). It was displayed for 0.3 seconds. There were 100 trials.

The grating's orientation was coded in such a way that at the beginning of each trial, it was randomly assigned either left or right tilt.

A loop was added in this routine where the highest amount of tilt was 20 degrees and lowest amount was 1 degree. These tilt degrees were adjusted based on the participant's responses using various step sizes such as 2, 1 and 0.5.

The participant displayed his/her responses using the 'right' arrow key [→] for a right tilt and the 'left' key [←] for a left tilt.

The data and responses of the participant was collected in an Excel sheet and was then analysed (Figure 2).

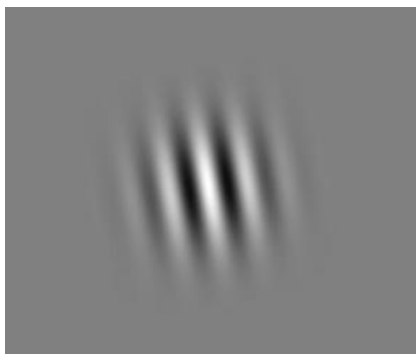


Figure 1

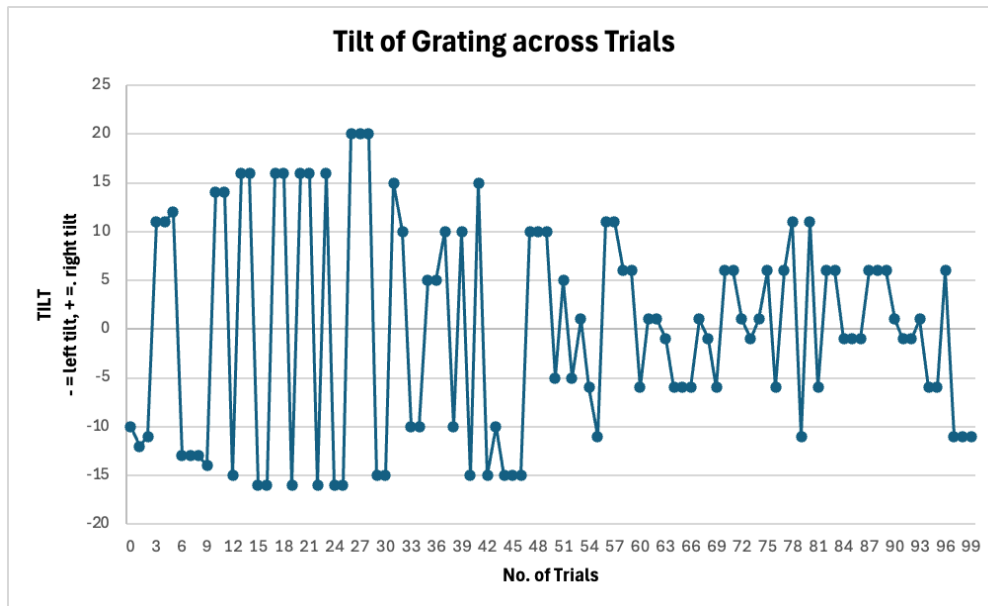


Figure 2

Results

The difference threshold was 1.6. This means that at this level, the participant was able to detect a change in the stimulus. Across all the 100 trials, the participant achieved an overall accuracy of 81%. The results showed a consistent pattern of performance. As seen in Figure 3, the correct and incorrect responses form the adaptive staircase.

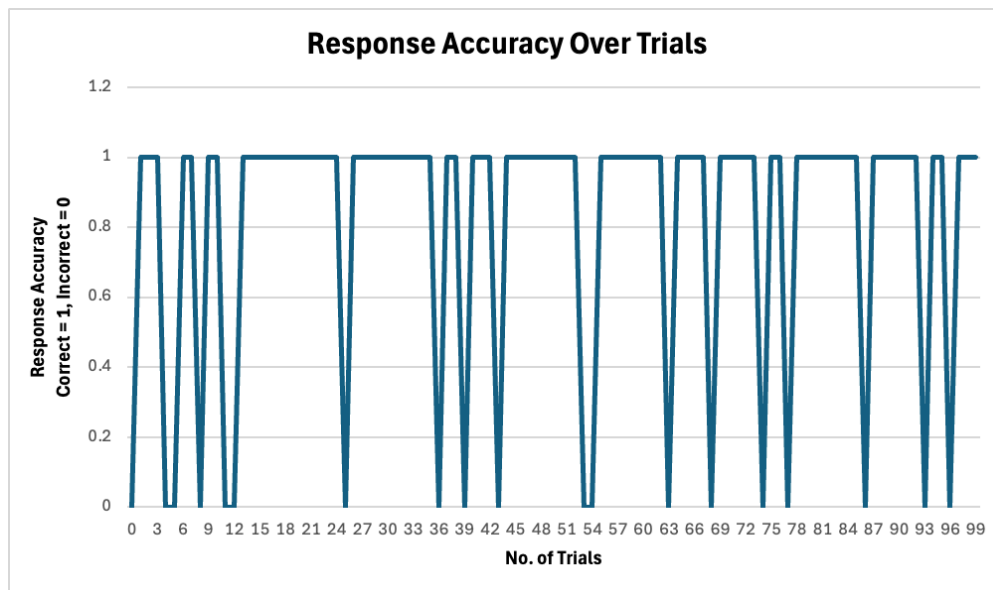


Figure 3

Discussion

The analysis of the participant's performance had a threshold of 1.6 degrees with an accuracy of 81%. The lowest tilt was 1 and the participant was able to detect the stimulus correctly. This shows that the participant was able to detect small stimuli as well and had a good stimulus response. The experiment may be unreliable in other conditions. In this setup, it was conducted where there were no external stimuli present to get the best results. Also, there was only one participant and thus, the results can't be generalized. A more reliable experiment would be where there is more than one participant.

References

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